EARTH SYSTEM COMPONENTS
12-GEO-M-AG02

Lectures: Miguel D. Mahecha

Exercises: Milena Mönks

Remote Sensing Centre for Earth System Research - rsc4earth.de



UNIVERSITÄT
LEIPZIG

WHO ARE WE?

Lectures: Miguel D. Mahecha



Miguel D. Mahecha

- A-level — Hannover, 1998
- Civil service — Chile, 1998 - 2000
- Studies — Geoecology, Universities of Bayreuth & Exeter, 2000 - 2006
- PhD — Land-atmosphere interactions, Max Planck Inst & ETH Zürich 2006-2009
- PostDoc — Climate extremes and CO_2, Max Planck Inst for Biogeochemistry 2009-2012
- Group Leader — Empirical Inference in the Earth System, 2012-2020
- Professor — Environmental Data Science and Remote Sensing, Leipzig University

Exercises: David Montero



David Montero

- BEng — 2011-2016 BEng in Topographic Engineering (Geomatics + RS), Cali, Colombia
- MSc — 2019-2020 MSc in Data Science, Santander, Spain
- PhD — 2021-2026 PhD in Physics and Earth System Science, Leipzig University
- Volunteer — 2022-2026 Google Developer Expert (GDE) for Google Earth Engine (GEE)
  2026-ongoing Google Earth Engine (GEE) Insiders Group
- Developer — Developer of free and open-source software for geospatial (FOSS4G)
  Awesome Spectral Indices, cubo, sen2nbar, eemont, etc.

Exercises: Milena Mönks



Milena Mönks

- 2011 A-level, research trip to Norway on Northern lights
- 2011-2012 Voluntary Ecological Year in Peru/Brasil
- 2012-2016 B.Sc. in Physics in Münster and Lisbon
- 2016-2019 M.Sc. in Environmental Sciences/Cluster Environmental Physics in Greifswald
- 2019 Research fellow at the German Aerospace Center (DLR), Oberpfaffenhofen, Team Smart Cities and Spatial Development/ Urban remote sensing
- since 2020 Research fellow in Environmental Data Science, Leipzig University

WHO ARE YOU? ...

Welcome to the University!

Welcome to the right subject!

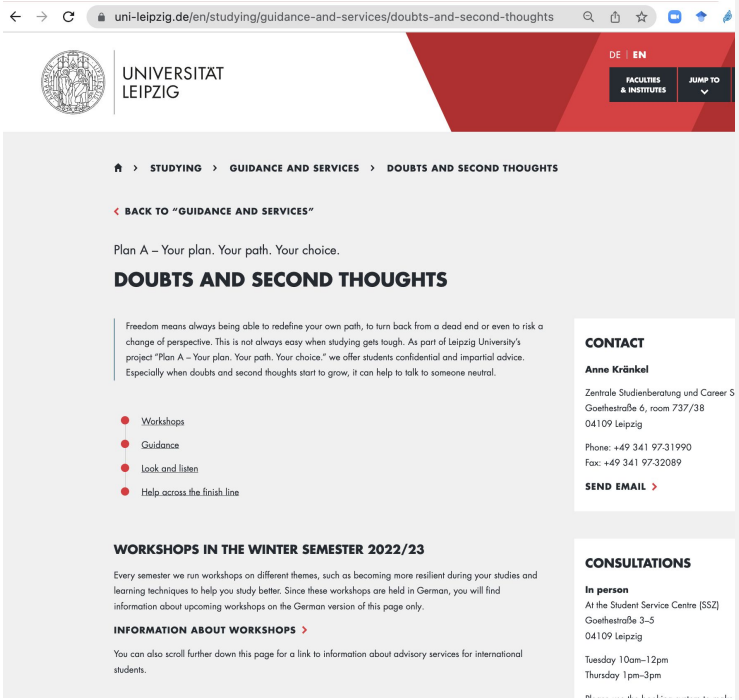
What we expect: curiosity and self-organisation

- Curiosity & self-motivation! We operate at eye-level
- Self-directed learning → Self-organisation
- Few guidelines, but recommendations → You decide with which one you can learn well ..

Starting is hard – look for contact, form groups

Starting new studies is always hard – no question about it!

- Look for contact!
- Form groups!
- Seek help when you need it - also from us! We might not be the person you need, but we help you find the person who can really help you!

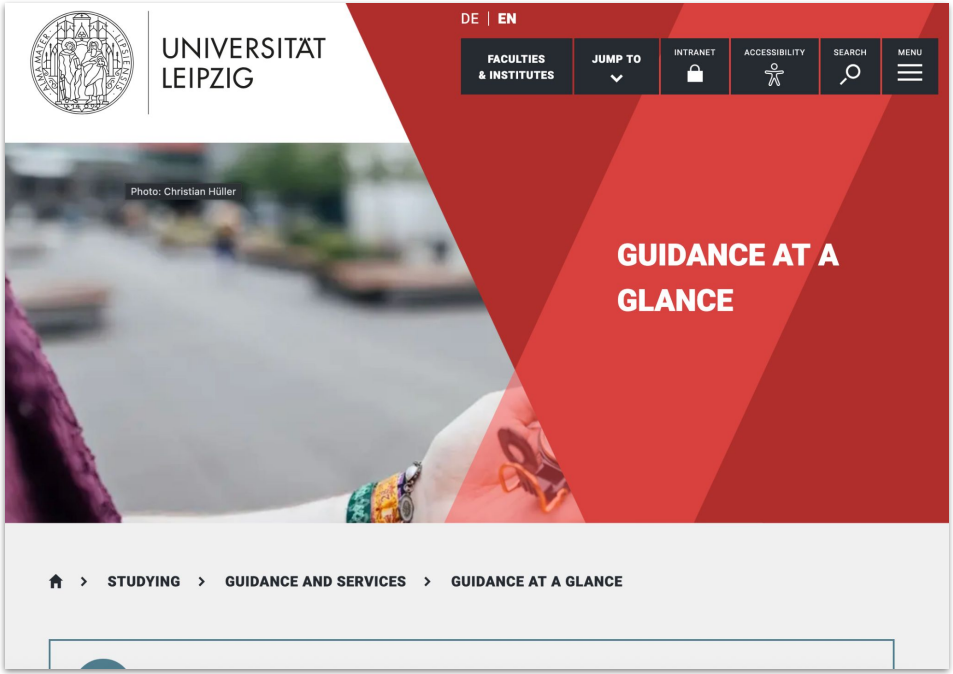


Support structures at the university

Guidance at a glance

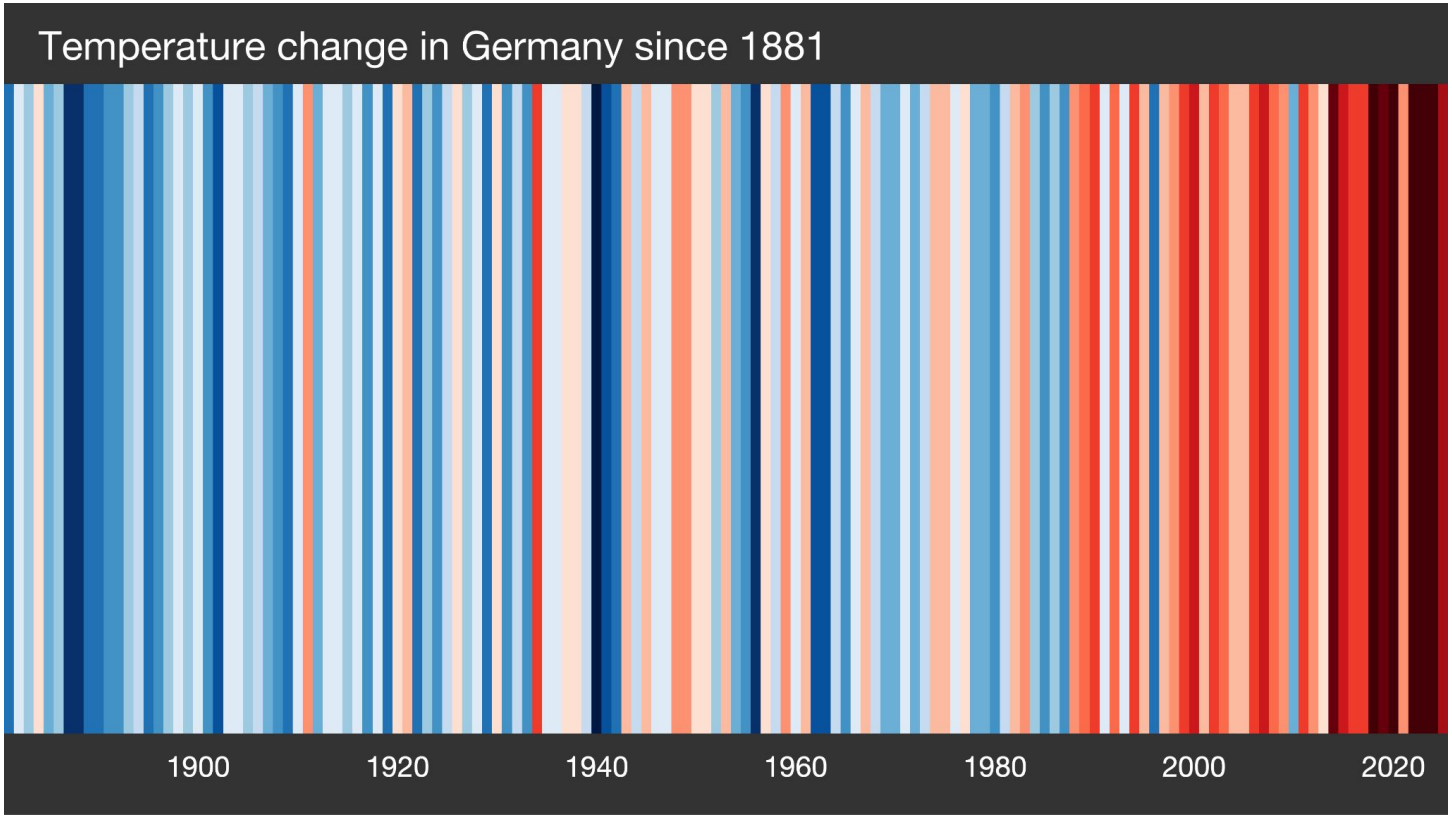
Advice available for all stages

- Before, during and after your studies
- Financing
- Accommodation
- Family responsibilities
- Health Impairment
- Psychosocial Counselling
- ...



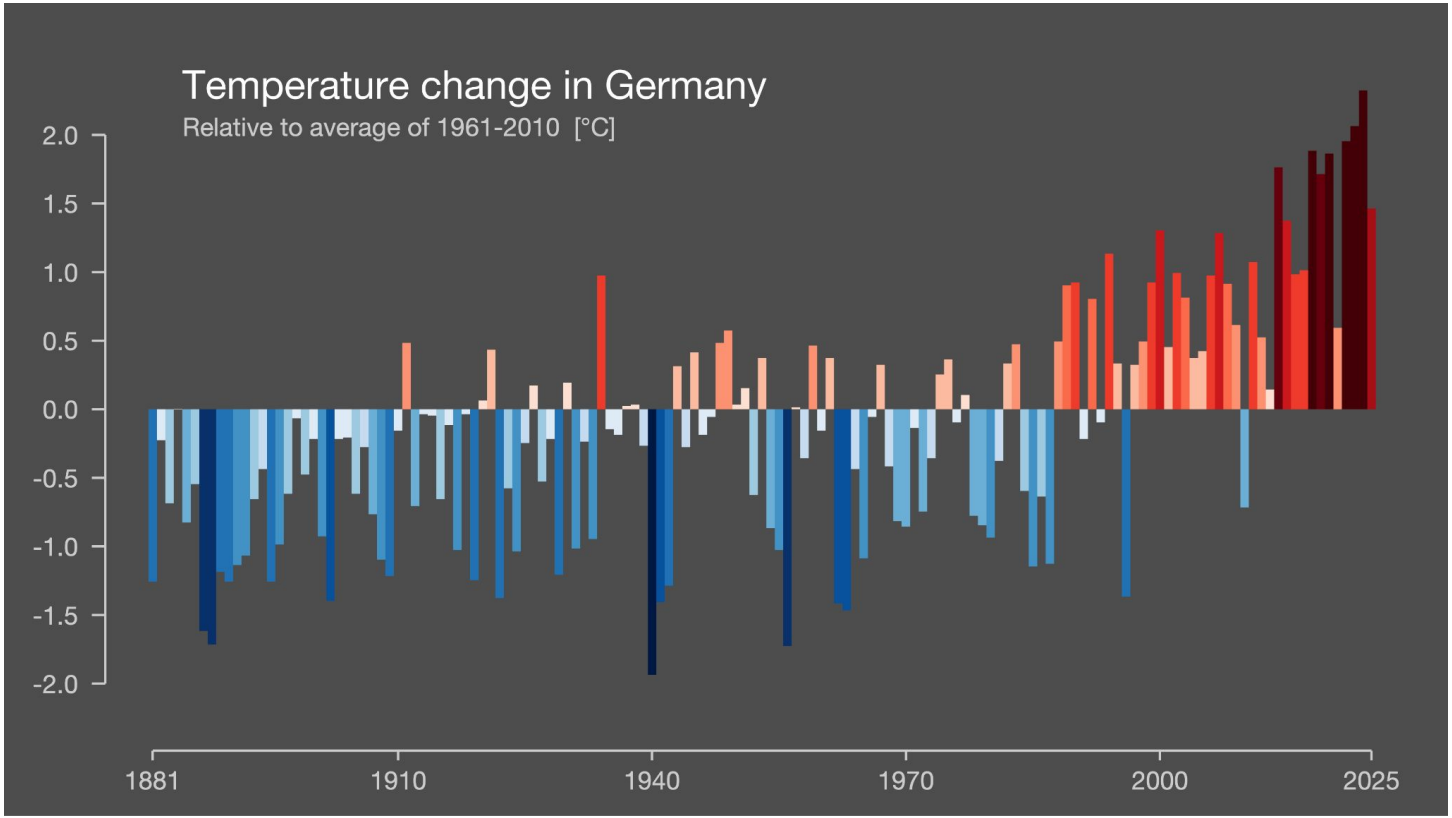
ONE POSSIBLE MOTIVATION

What does climate change look like?



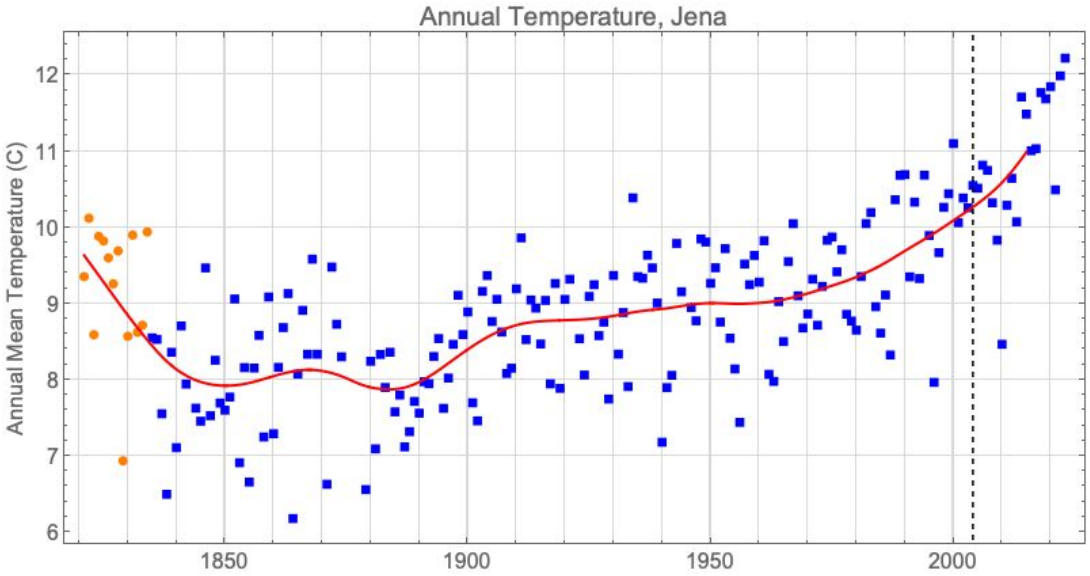
Graphic: #ShowYourStripes – Ed Hawkins, University of Reading (showyourstripes.info)

Warm years have piled up since the 1990s



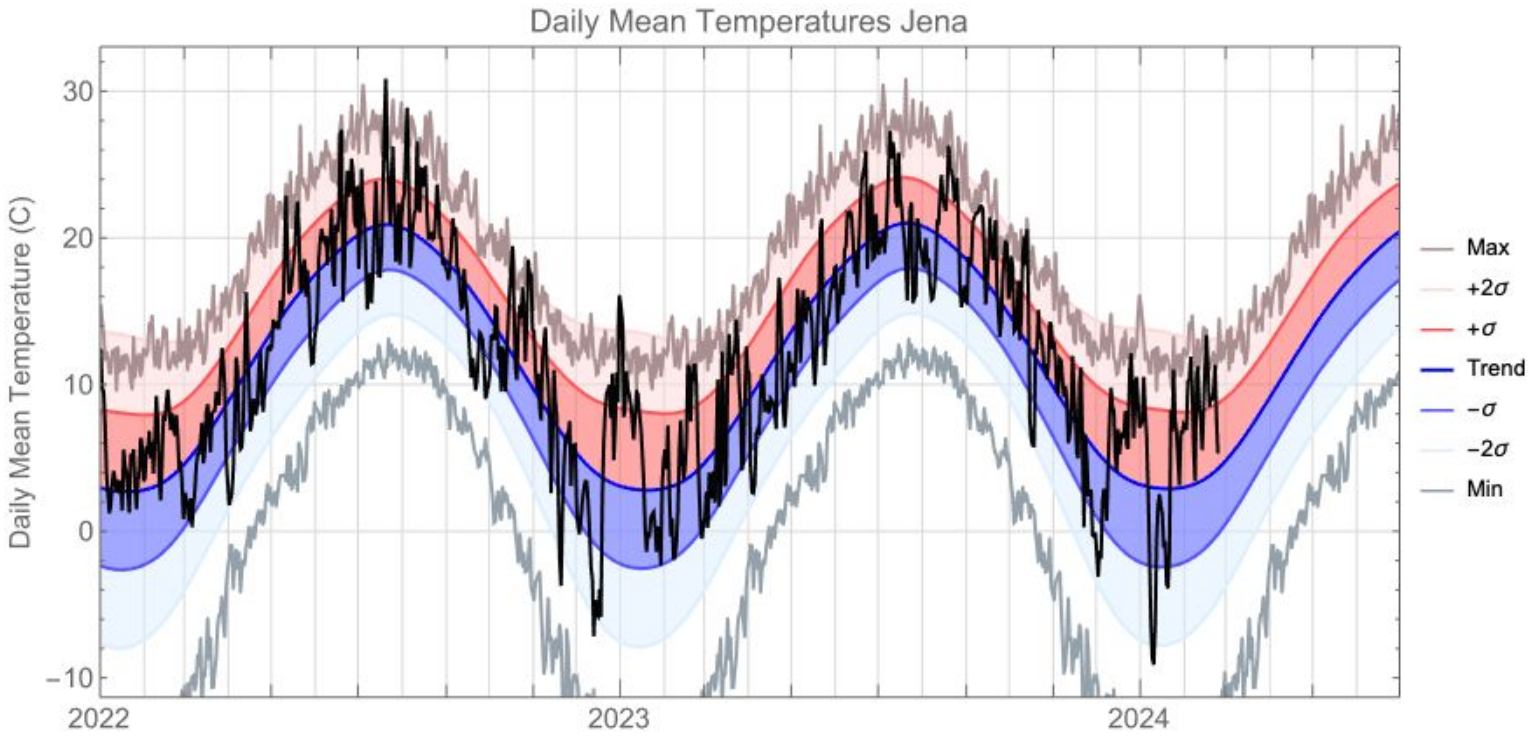
Graphic: #ShowYourStripes – Ed Hawkins, University of Reading (showyourstripes.info)

Can we measure climate change?



Annual mean temperatures in Jena since 1821. Analysis and figures: Prof. Dr Martin Heimann;
https://www.bgc-jena.mpg.de/~martin.heimann/weather/weather_temperature/
Orange points (left): different measurement conditions – not comparable with the rest of the series.

Temperatures in my hometown Jena …



- Analysis: Prof. Martin Heimann

Nobel Prize in Physics 2021 – for what?

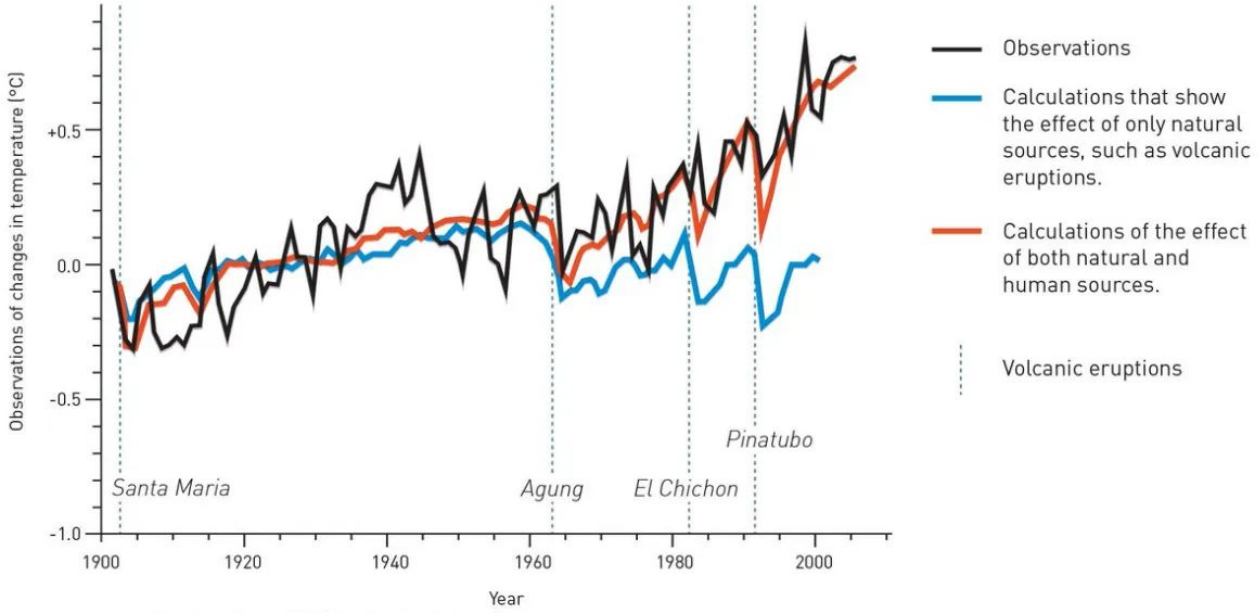
Nobel prize 2021 in physics: Climate Attribution



Nobel Prize 2021: fingerprints of human influence

Identifying fingerprints in the climate

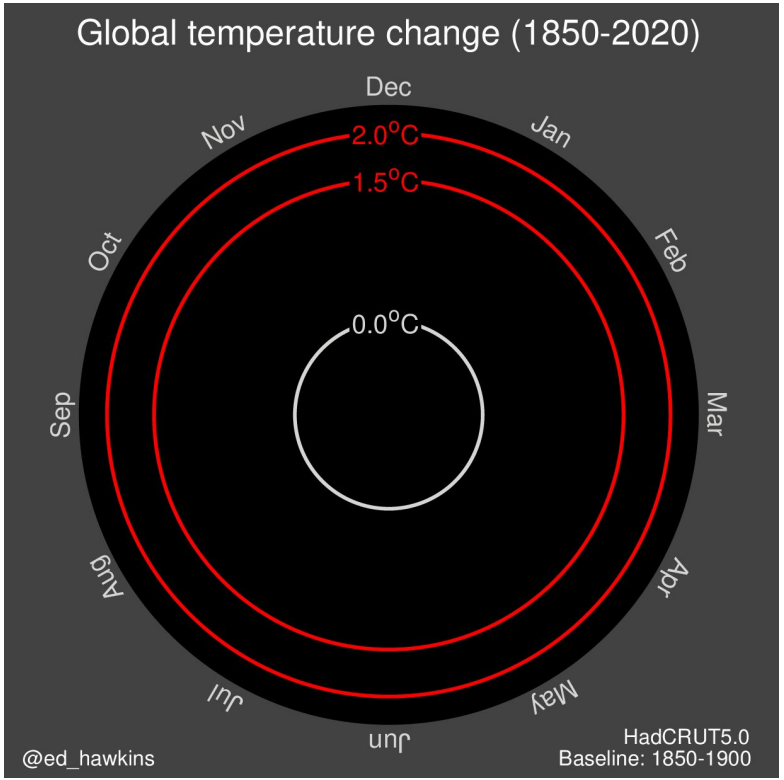
Klaus Hasselmann developed methods for distinguishing between natural and human causes (fingerprints) of atmospheric heating. Comparison between changes in the mean temperature in relation to the average for 1901–1950 (°C).



Source: Hegerl and Zweirs (2011) Use of models in detection & attribution of climate change, *WIREs Climate Change*.

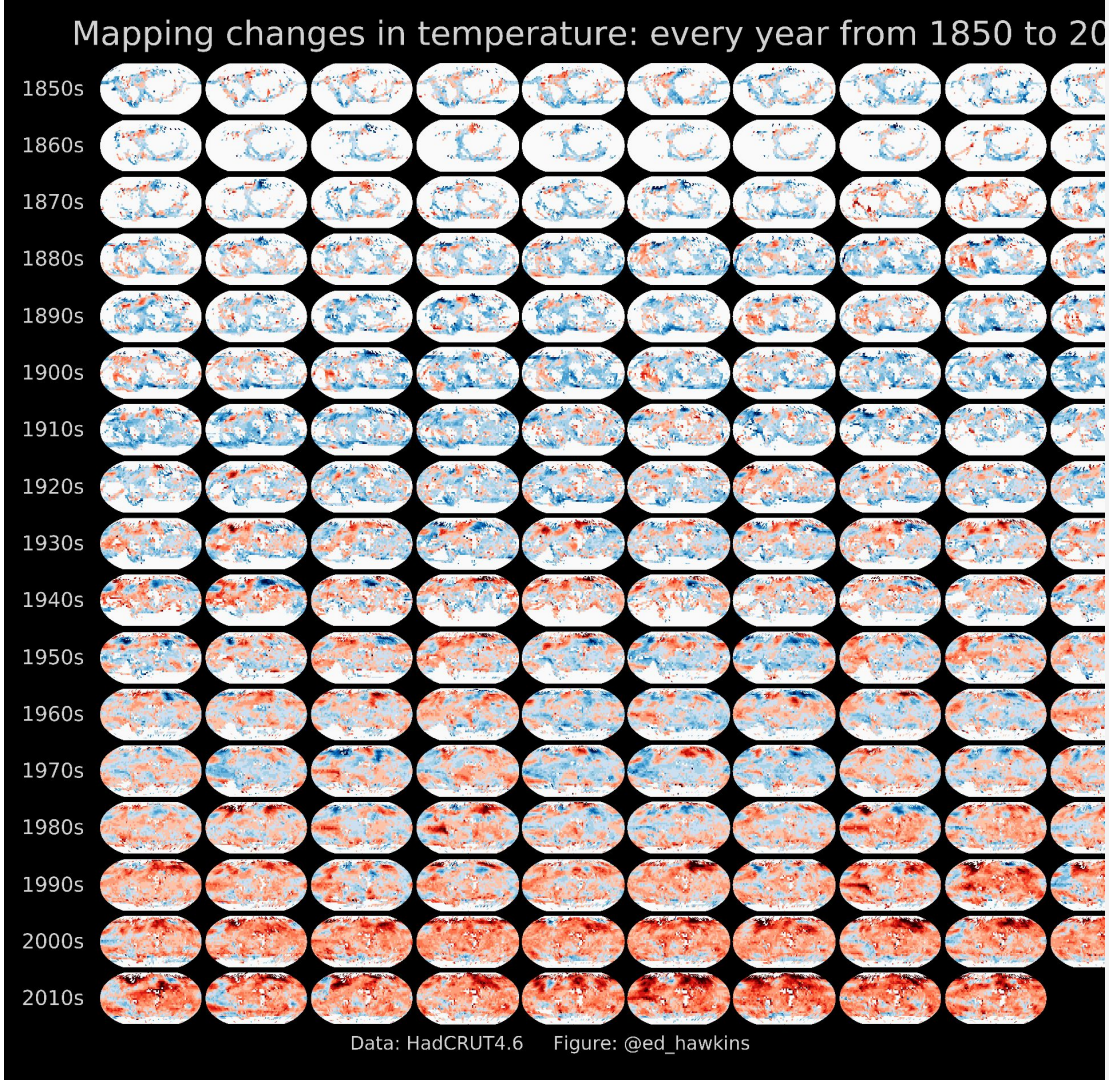
©Johan Jarnestad/The Royal Swedish Academy of Sciences

Why does everyone worry about +1.5 °C?



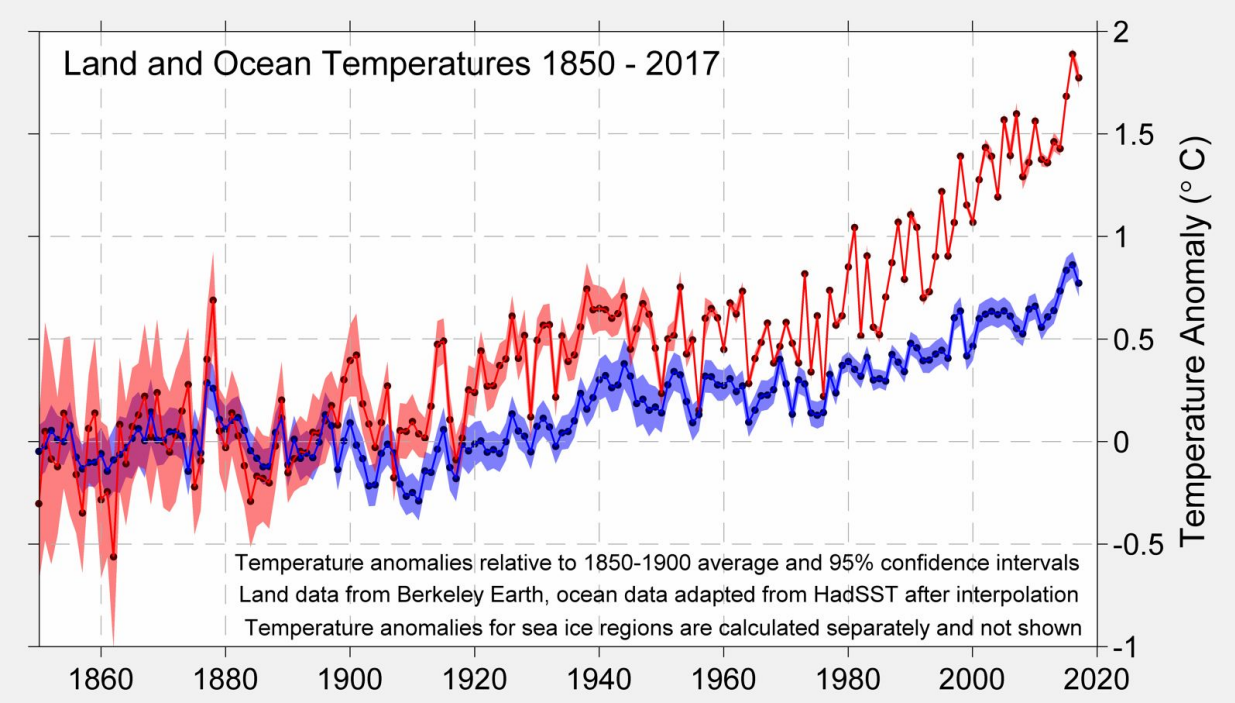
http://www.climate-lab-book.ac.uk/

Is the world warming equally everywhere?



https://www.climate-lab-book.ac.uk/2016/mapping-global-temperature-change/

Can you explain this?



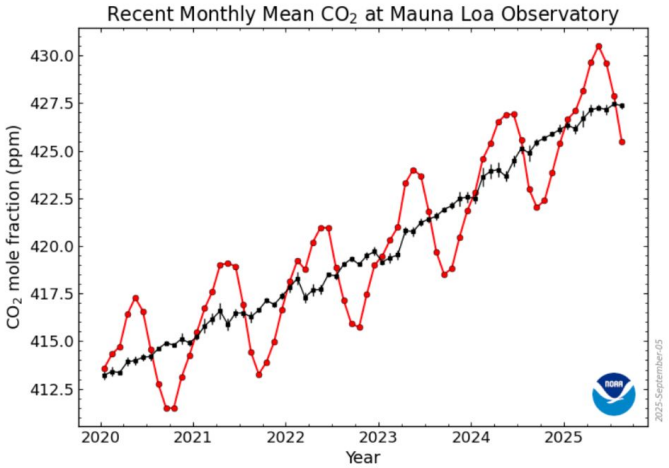
http://berkeleyearth.org/global-temperatures-2017/

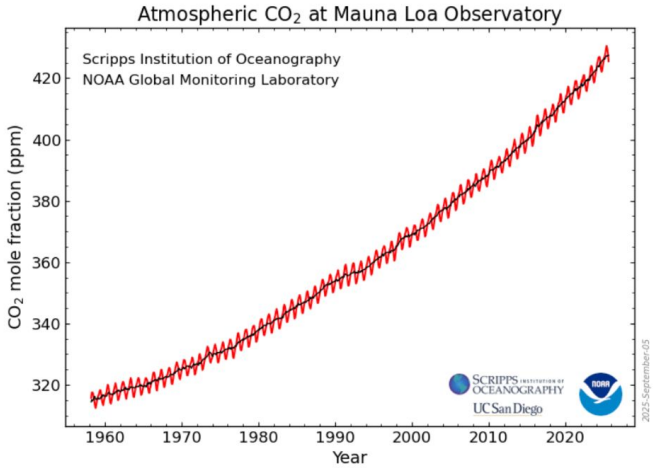
Is this the only cause? What are the interactions triggered her

Last Month | Last 1 Year | Full Record | Growth Rate | Data | Interactive Plots

Monthly Average Mauna Loa CO_2

August 2025: 425.48 ppm
August 2024: 422.99 ppm
Last updated: Sep 05, 2025





Is this cool? Warmer is nicer and so?



What do people perceive?



Niedersachsen

Jetzt sogar 42,6 Grad - deutscher Hitzerekord im Emsland

Der Deutsche Wetterdienst hat im niedersächsischen Lingen mehr als 42 Grad gemessen. Noch ist der Rekord nicht offiziell bestätigt, doch es wäre der höchste Wert seit Aufzeichnung der Temperaturdaten in Deutschland.



Wetter

MITTEN IN NRW: DREI TAGE AM STÜCK ÜBER 40 GRAD!

27.07.2019 07:20 | 463

ERSTMALS IN DEUTSCHLAND DREI TAGE MIT MEHR ALS 40 GRAD VERZEICHNET



Hitzerekord

Niederlande erstmals heißer als 40 Grad

25. JULI 2019 UM 16:50 UHR | Lesedauer: Eine Minute

Note: the 42.6 °C measured at Lingen (Emsland) on 25 July 2019 was later annulled by the German Weather Service (DWD, December 2020). The official German record is 41.2 °C (Duisburg-Baerl and Tönisvorst, 25 July 2019).

Science and society …



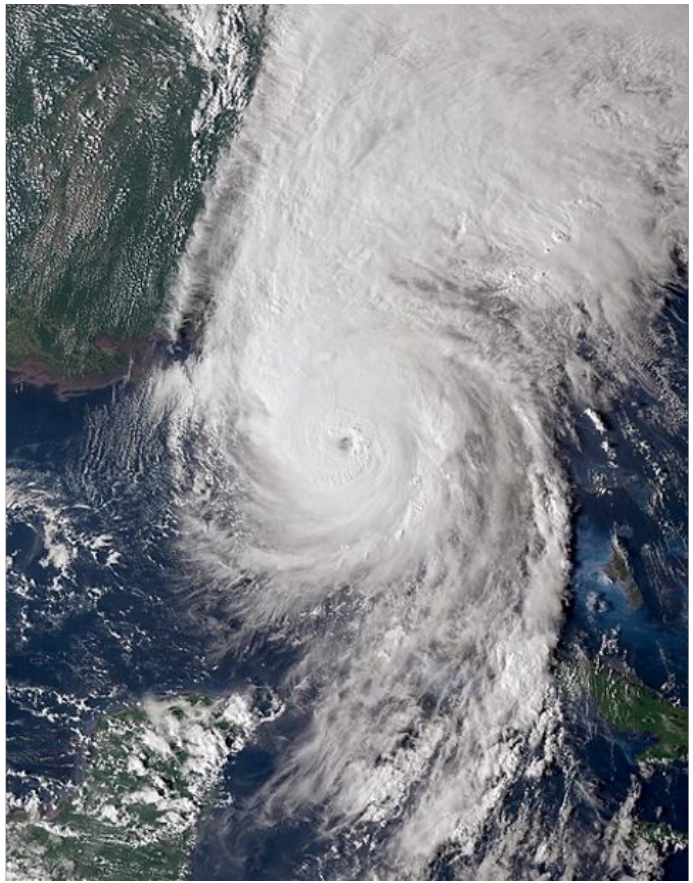
on | Sport | Culture | Lifestyle | The Guardian

e Environment Science Global development Football Tech Business Obituaries

'It's mindblowing': US meteorologists face death threats as hurricane conspiracies surge

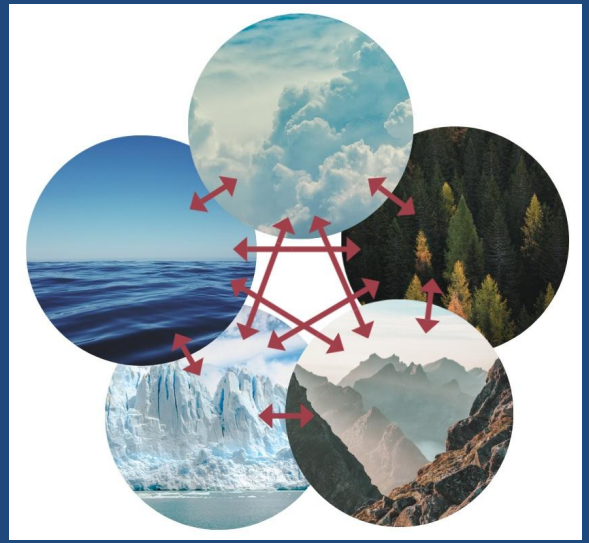
Storms Helene and Milton have triggered rise of misinformation stoked by Trump and fellow Republicans

Two couples who evacuated get a ride back to their home through flooding on Thursday after Hurricane Milton hit the region in Palm Harbor, Florida. Photograph: Mike Carlson/AP



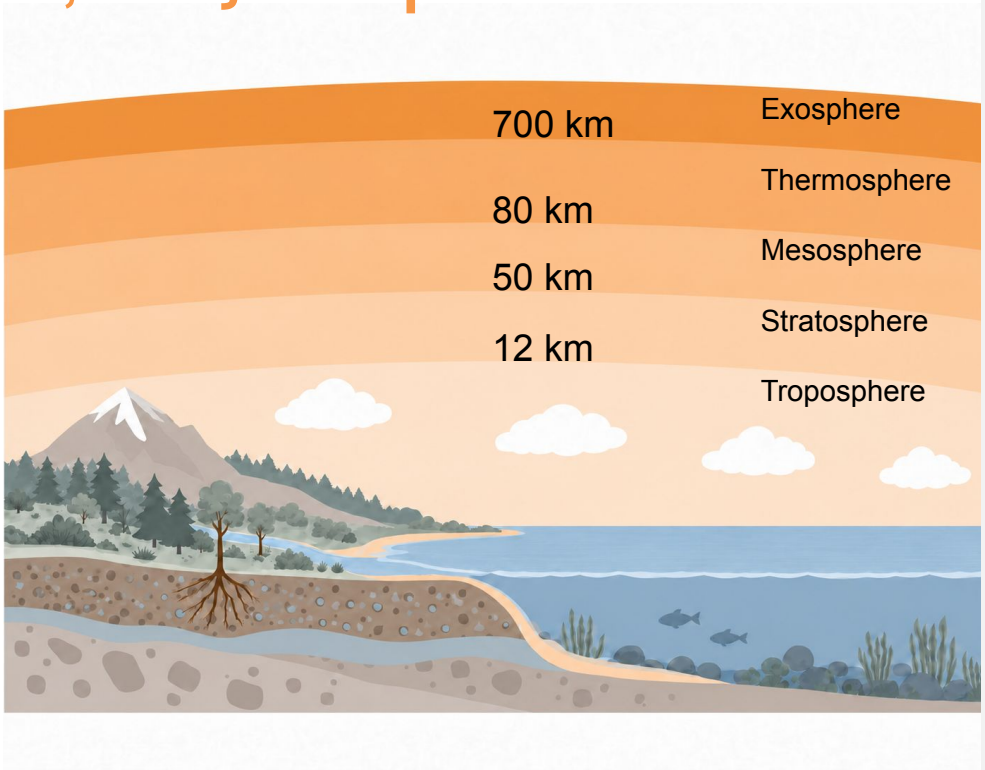
WHAT IS YOUR MOTIVATION?

ONE MOTIVE: UNDERSTANDING THE EARTH SYSTEM



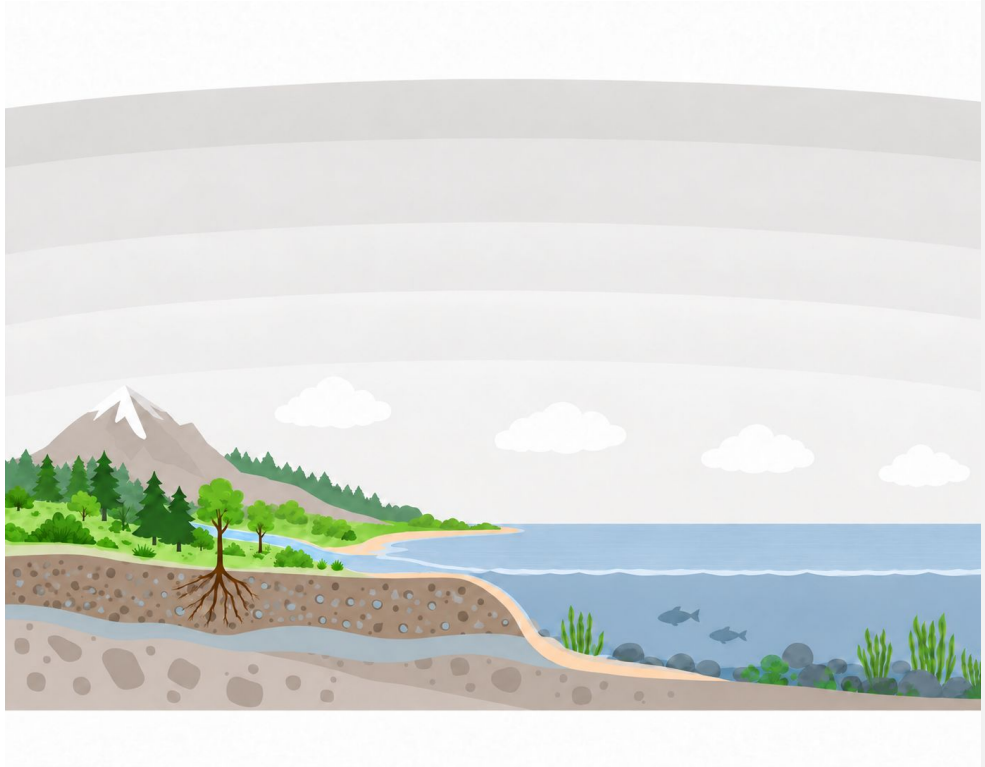
The atmosphere: one thin layer, many disciplines

- Composition of the atmosphere
- Physics of atmospheric processes
- Climate history
- Regional phenomena
- Meteorology
- Micrometeorology
- Climate protection

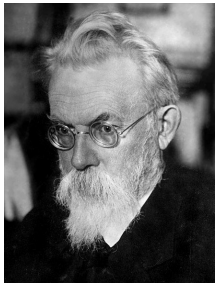


The biosphere

- Thin, life-supporting layer at the Earth's surface.
- Upper limit: atmosphere
- Lower limit: deep-sea vents
- All living organisms and their environment: an estimated 8.7 million eukaryotic species (Mora et al., 2011) – plants, animals, fungi and unicellular eukaryotes such as protozoa – plus prokaryotes such as bacteria and archaea.

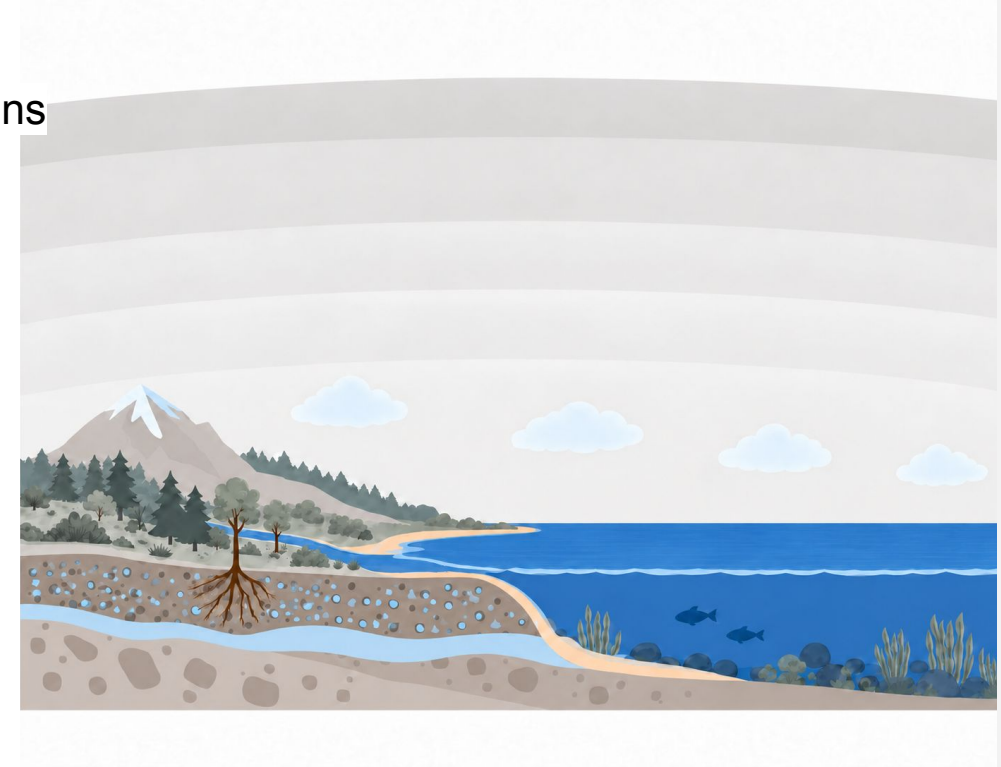


Konzept: Wladimir Wernadski



The hydrosphere

- Oceanography
  - State and dynamics of the oceans
- Limnology
  - Lake research
- Classical hydrology
  - Runoff dynamics
  - Storage dynamics
  - Soil water dynamics
- Water in the atmosphere
  - atmospheric water vapour
  - Water clouds and ice clouds



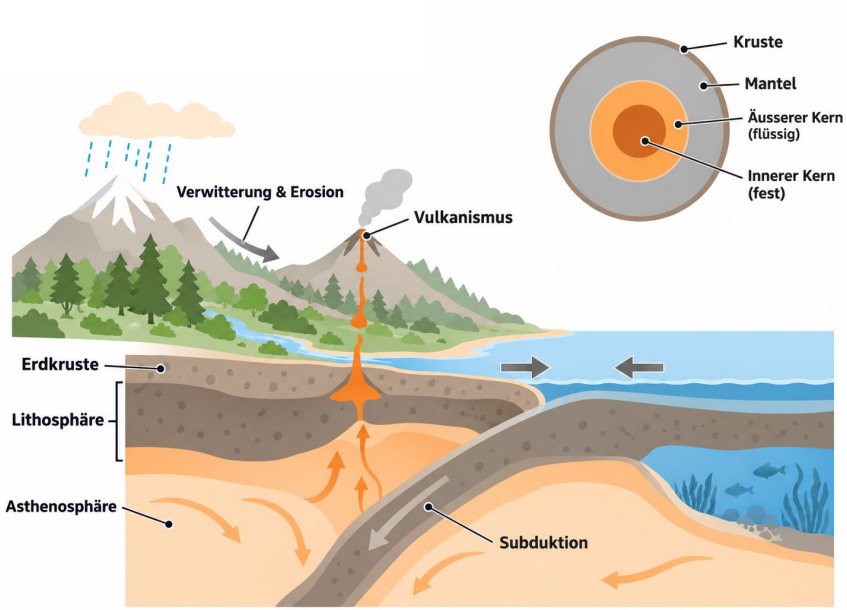
The geosphere

- Geology, geophysics etc.

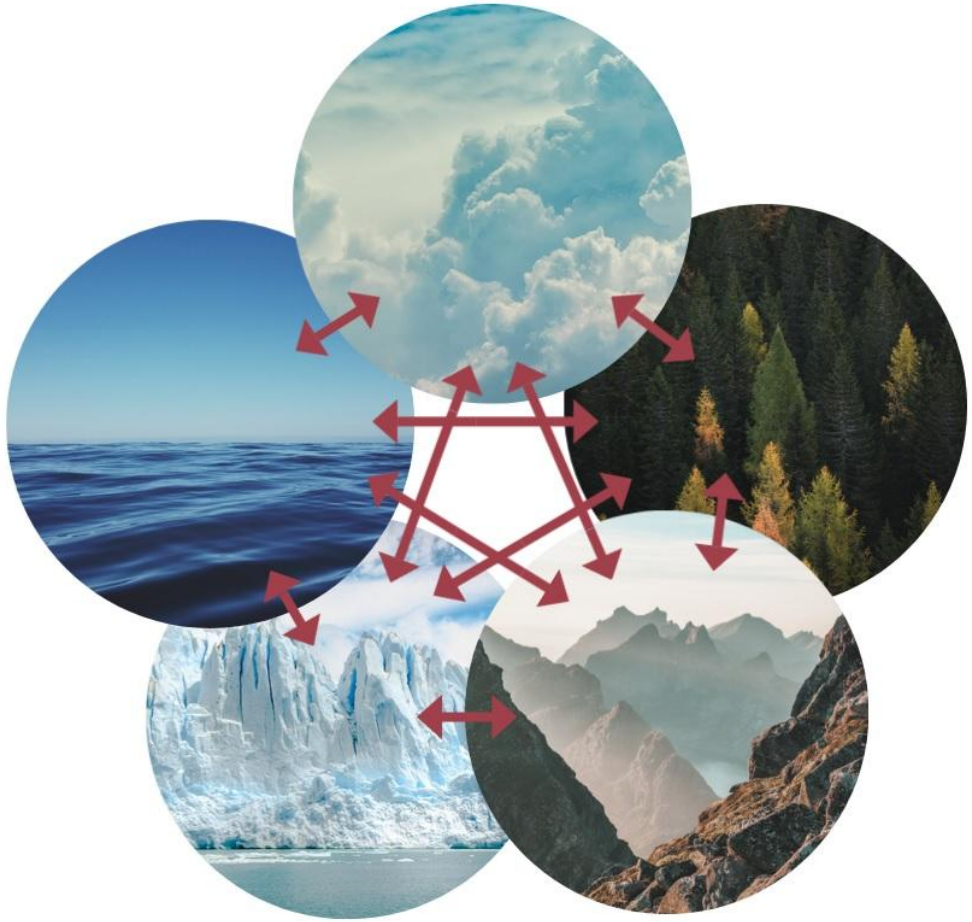
- Endogenous dynamics:
  - Internal structure of the Earth system
  - Understanding the structure of the Earth's crust
  - Formation of rocks, deformation
  - Seismic activity

- Exogenous dynamics:
  - Geomorphology
  - Weathering, erosion, (soil formation)
  - Landforms

- Historical geology:
  - Reconstruction of the history of the Earth system



Wie hängen diese Sphären voneinander ab?



How do these spheres interact?



Fig. from: https://wiki.seg.org/wiki/Earth_system_science

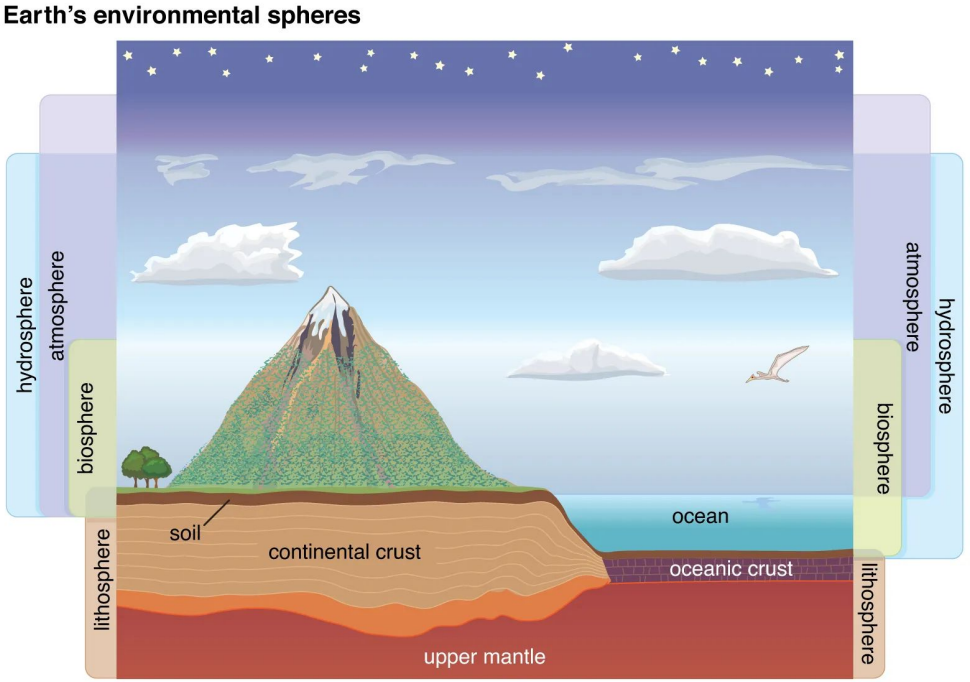
An important concept: the coupled Earth system

Hydrosphere → Atmosphere, Biosphere, Geosphere?

Atmosphere → Hydrosphere, Biosphere, Geosphere?

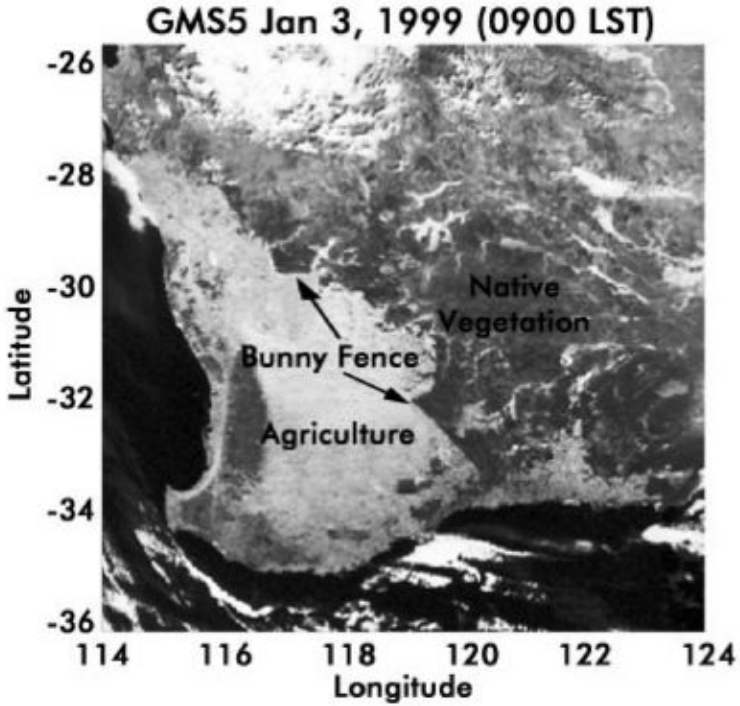
Biosphere → Hydrosphere, Atmosphere, Geosphere?

Geosphere → Hydrosphere, Atmosphere, Biosphere?



Biosphere → Atmosphere





Lyons, T.J. (2002) Clouds prefer native vegetation. Meteorology and Atmospheric Physics, 80, 131–140.

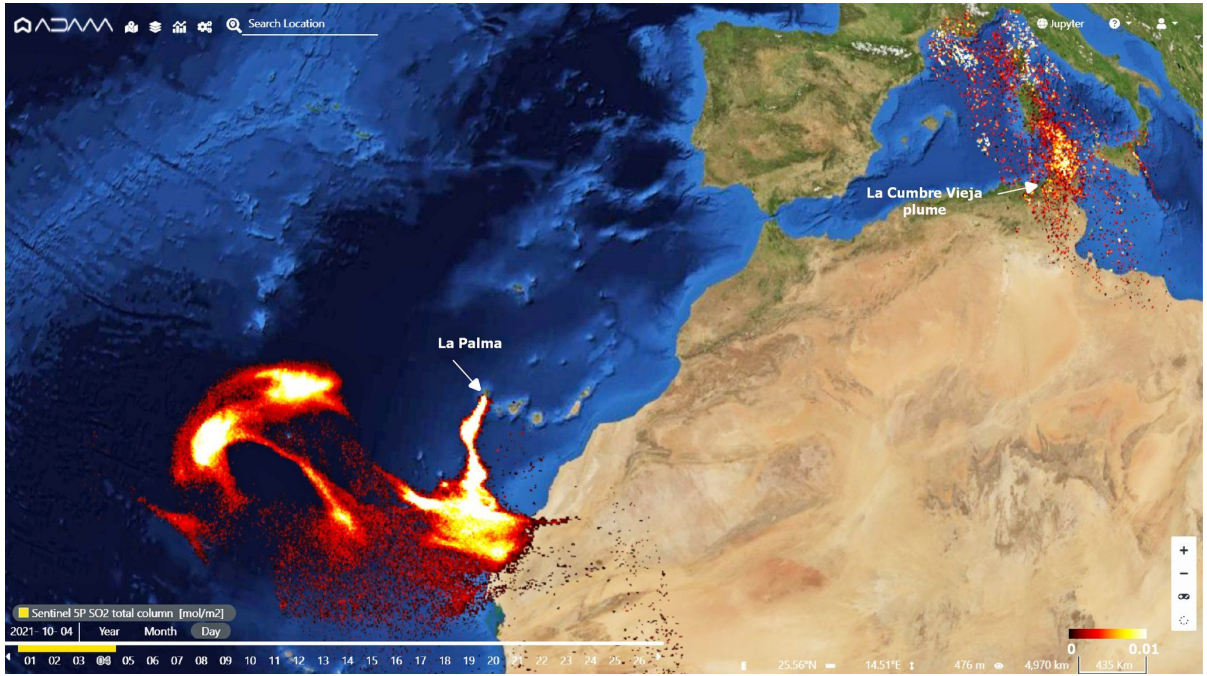
Geosphere → Atmosphere



Gas-ash mixture rises, cannot rise further due to atmospheric stratification, but induces cloud formation which then spreads concentrically

01.10.2021
https://scitechdaily.com/satellite-captures-stunning-gravity-waves-over-la-palma-from-cumbre-vieja-eruption/

Geosphere → Atmosphere



SO2 is distributed vertically depending on atmospheric dynamics

04.10.2021
https://twitter.com/PlatformAdam/status/1445098357198921728

Geosphere → Atmosphere → Biosphere



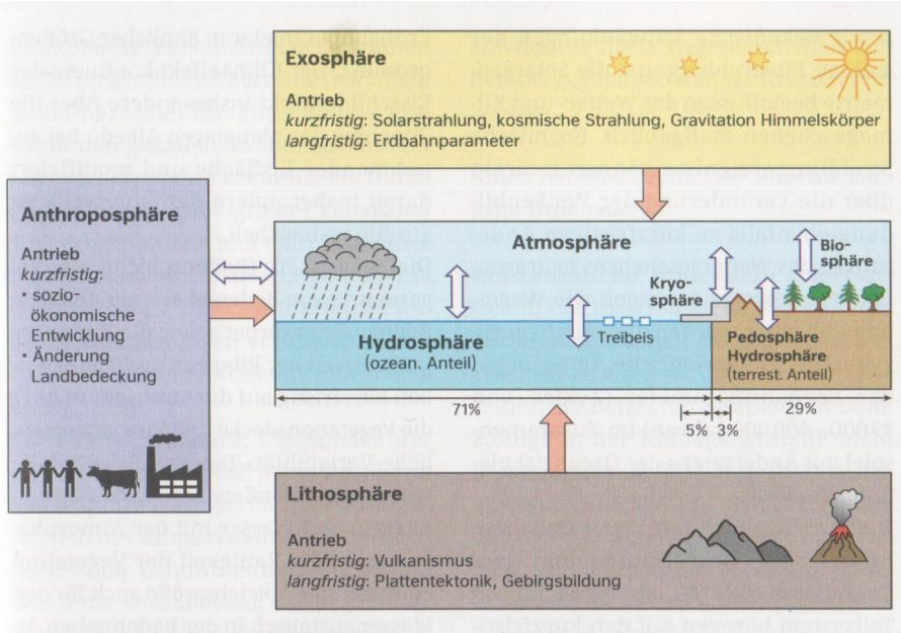
07.02.2021
https://www.lvz.de/Leipzig/Lokales/Blutschnee-Darum-faerbte-sich-der-Schnee-in-Leipzig-rot

How do these spheres interact? Exercise 1:

What is the (fraction of) mass of each sphere, in which the main interactions in contemporaneous Earth system dynamics take place?

What is the (fraction of) volume of each sphere, in which the main interactions in contemporaneous Earth system dynamics take place?

“Climate” is nothing independent!



Bendix & Luterbacher (2019) *Klimatologie*, Westermann, Braunschweig

Climate: an exchange system

"Climate can also be understood in simplified terms as a system whose task is to balance spatial energetic differences.

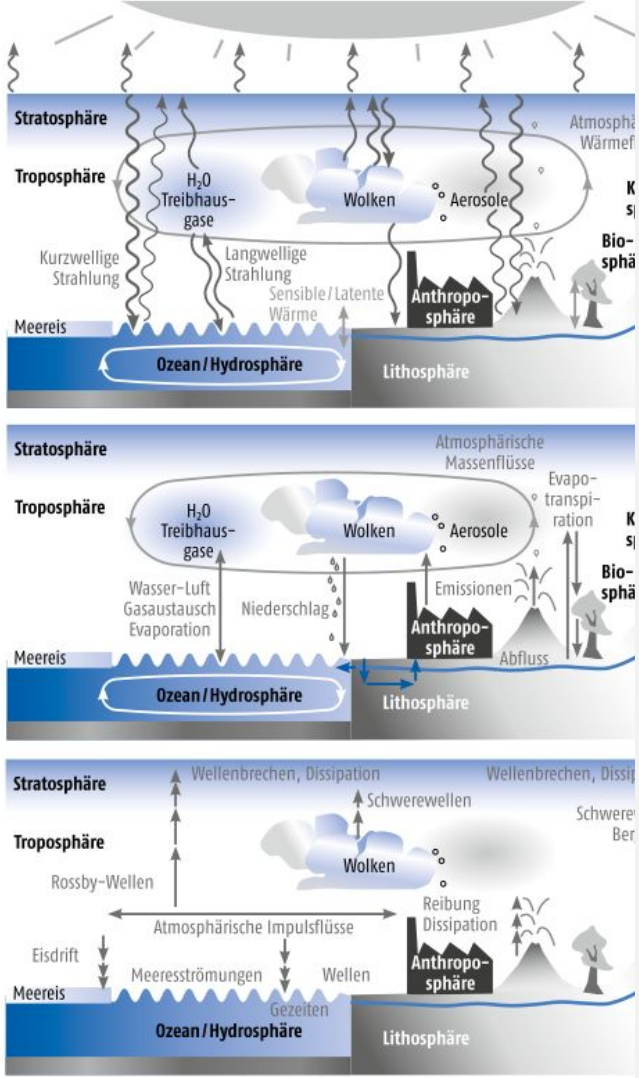
In the process, energy, mass and momentum are transported and exchanged horizontally and vertically.

The processes involved run on temporal and spatial scales that span several orders of magnitude."

Energy

Masses

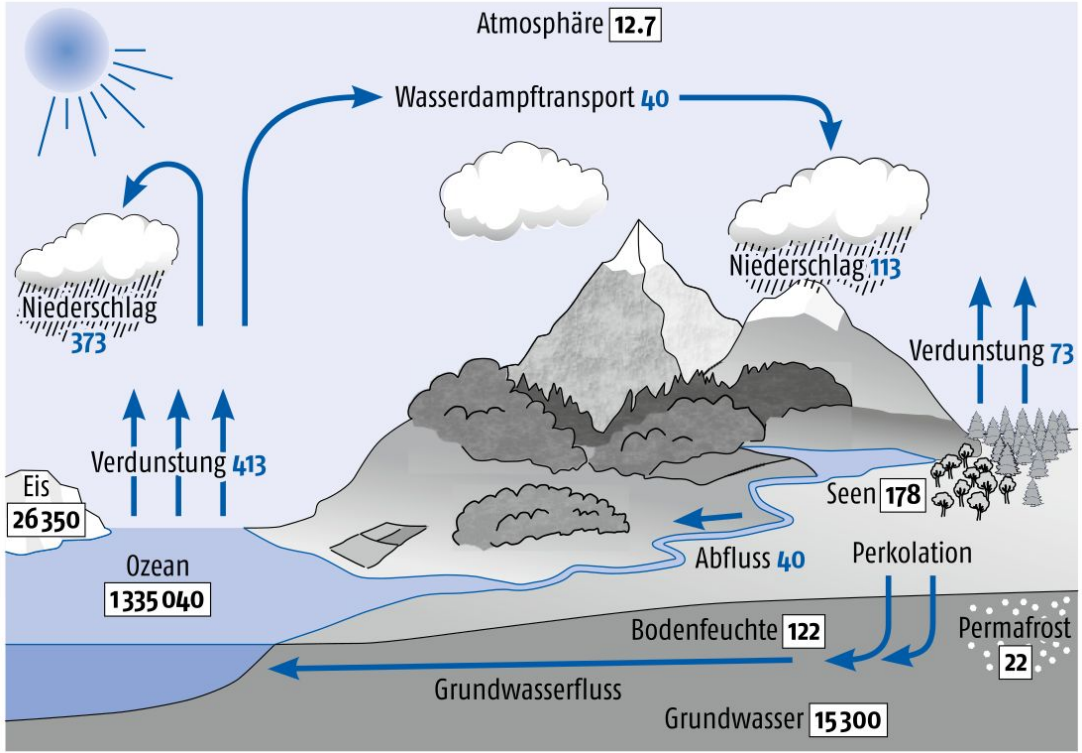
Momentum



Brönnimann (2018) *Klimatologie*, utb, Haupt Verlag, Berlin

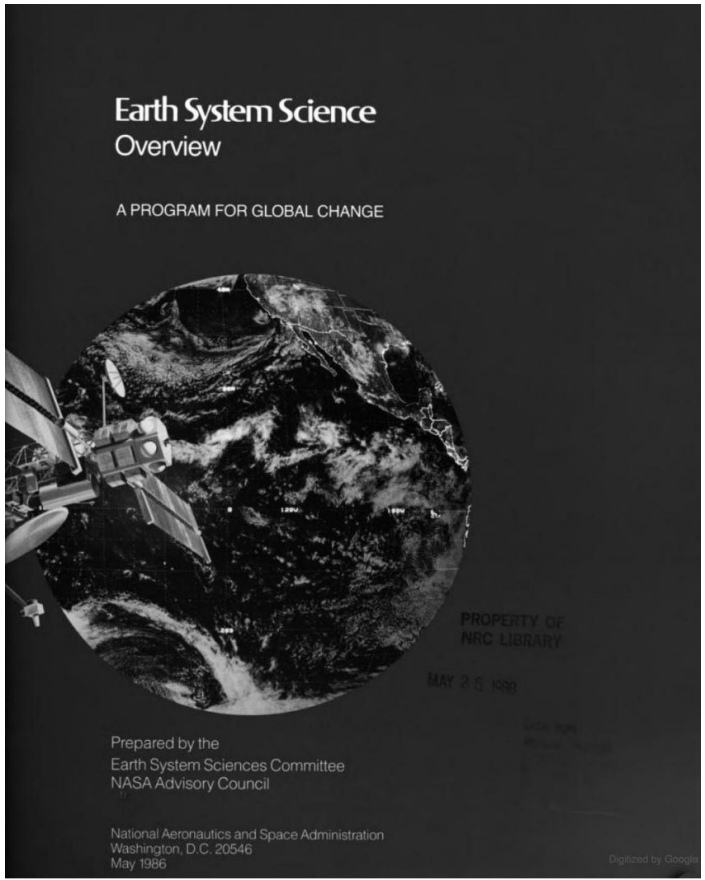
Each process is part of a system of exchange rates

"Reservoirs are given in 1000 km3, rivers in 1000 km3/year".



Brönnimann (2018) *Klimatologie*, utb, Haupt Verlag, Berlin

The Earth system – how we understand it here

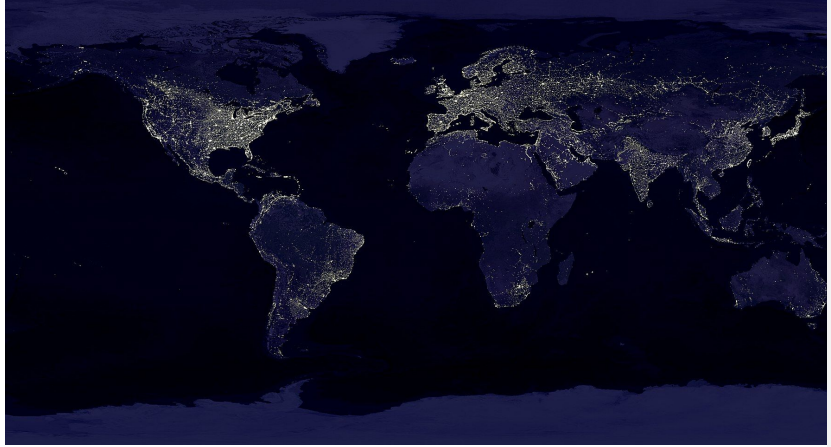


"The goal of Earth system science is to obtain a scientific understanding of the entire Earth system on a global scale by describing how its component parts and their interactions have evolved, how they function, and how they may be expected to continue to evolve on all timescales"

Earth System Sciences Committee NASA Advisory Council, 1986

Anthropocene: a human epoch?

"Anthropocene (anthropos = the erect man). The Anthropocene, which began around 1850, is the era in which humans are the main driving force of natural changes." (Brönnimann, 2018)



Translated from German. Note: Greek ánthrōpos means “human being”. In March 2024 the International Union of Geological Sciences (IUGS) rejected the proposal to define the Anthropocene as a formal epoch; it remains an informal term.

Paul Jozef Crutzen 3 December 1933 – 28 January 2021



concepts

Geology of mankind

Paul J. Crutzen

For the past three centuries, the effects of humans on the global environment have escalated. Because of these anthropogenic emissions of carbon dioxide, global climate may depart significantly from natural behaviour for many millennia to come. It seems appropriate to assign the term ‘Anthropocene’ to the present, in many ways human-dominated, geological epoch, referring to the “anthropozoic era”. And in 1926, V. I. Vernadsky acknowledged the increasing impact of mankind: “The direction in which the processes of evolution must proceed, namely towards increasing consciousness and thought, and forms having greater and greater influence on their surroundings.” Teilhard de Chardin and Vernadsky used the term ‘noösphere’ — the ‘world of thought’ — to mark the growing role of human brain-power in shaping its

The Anthropocene

The Anthropocene could be said to have started in the late eighteenth century, when analyses of air trapped in polar ice showed the beginning of growing global concentrations of carbon dioxide and methane.

ozone-destroying properties of the halogens have been studied since the mid-1970s. If it had turned out that chlorine behaved

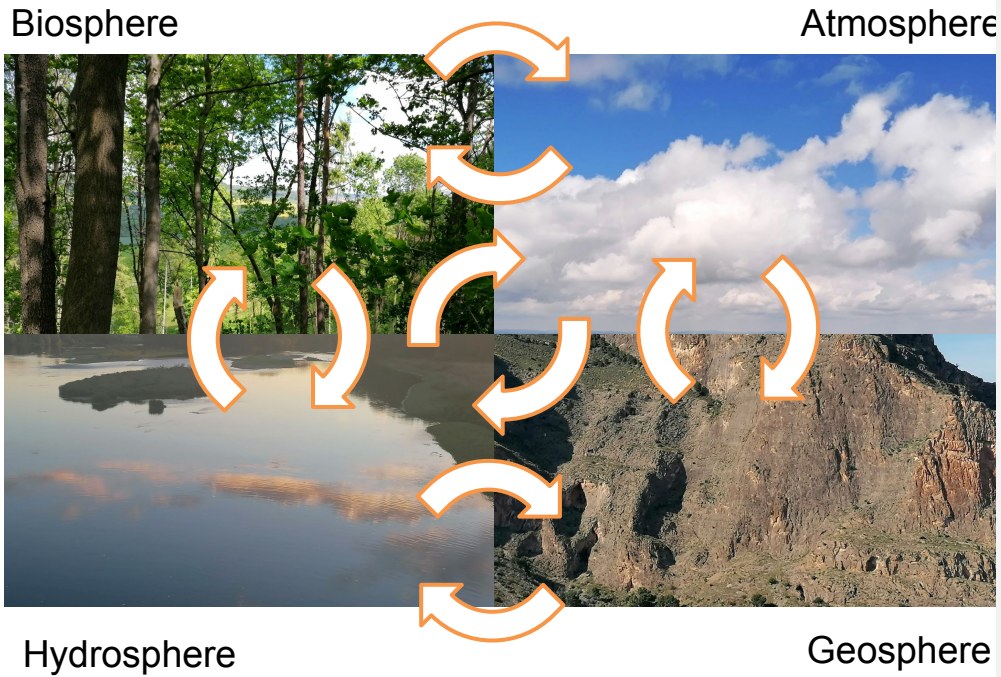
WHAT IS HAPPENING TODAY?



Global monitoring covers all spheres

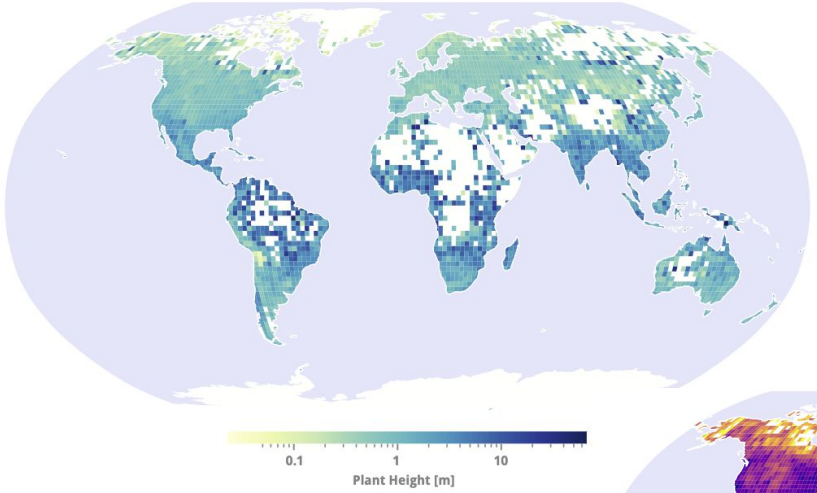


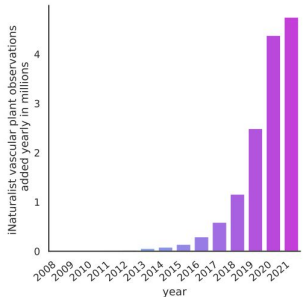
https://www.noaa.gov/

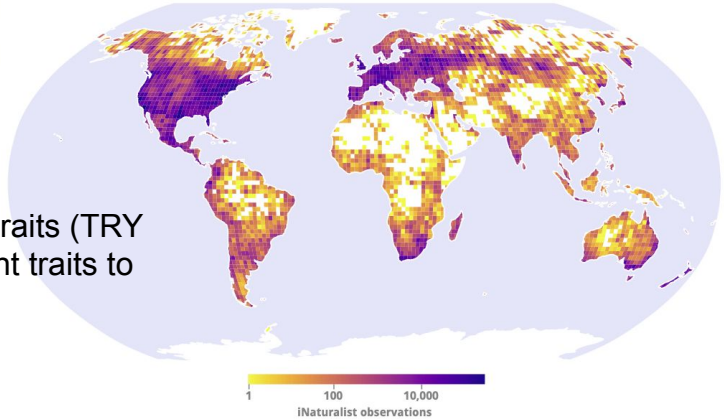


Citizen science → Global biodiversity patterns

Wolf, S., Mahecha, M. D., … & Kattenborn, T. (2022). Nature Ecology & Evolution.
https://doi.org/10.1038/s41559-022-01904-x





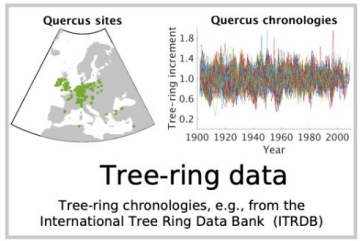


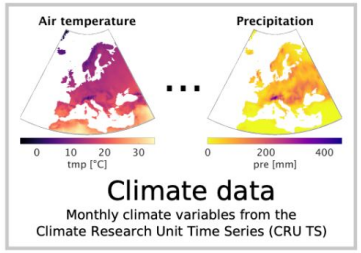
Linking iNaturalist observations to plant traits (TRY database) allows global maps of key plant traits to be estimated

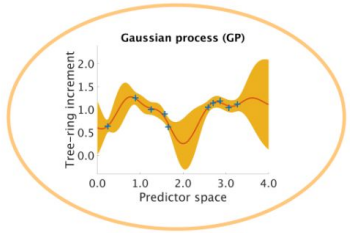


Point data → Spatiotemporal patterns



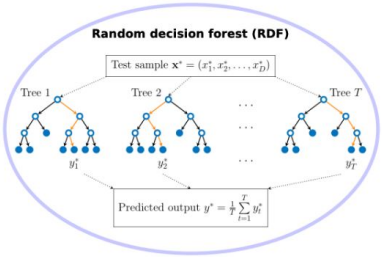


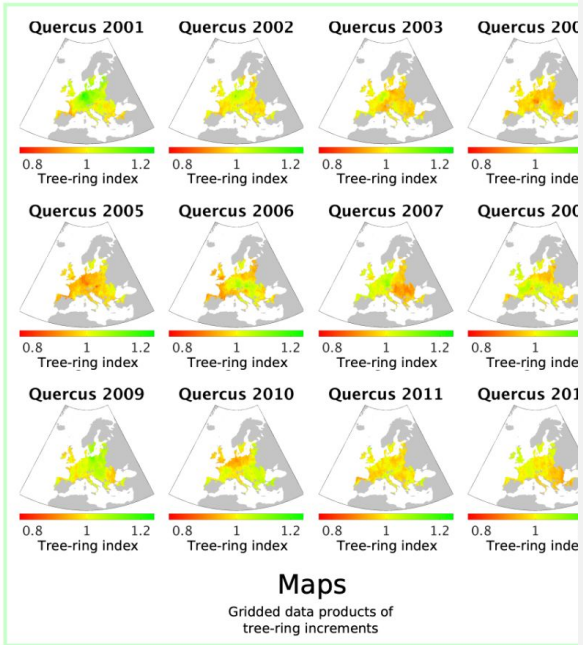




Machine learning

Regression algorithms

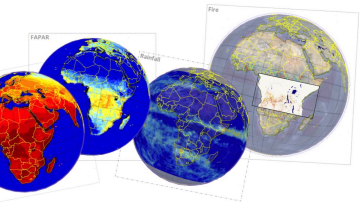




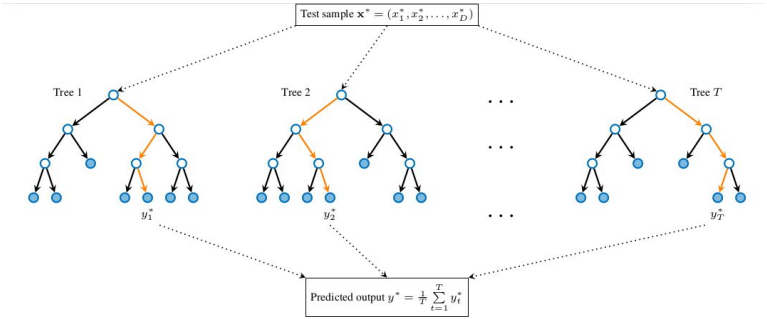
Bodesheim, P., … & Mahecha, M. D. (2022).
Environmental Data Science, https://doi.org/10.1017/eds.2022.8

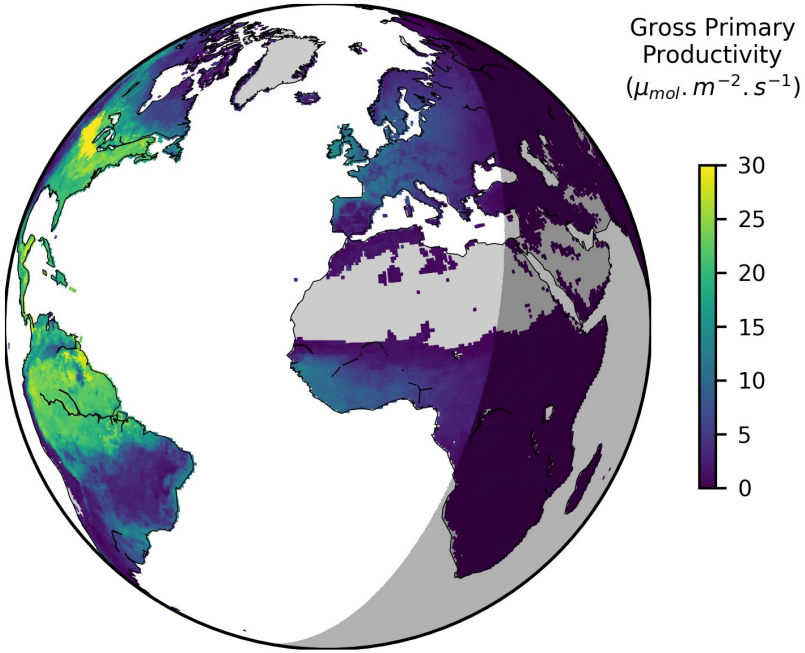
Unprecedented scales

GPP: Gross Primary Productivity =
Total uptake of CO_2





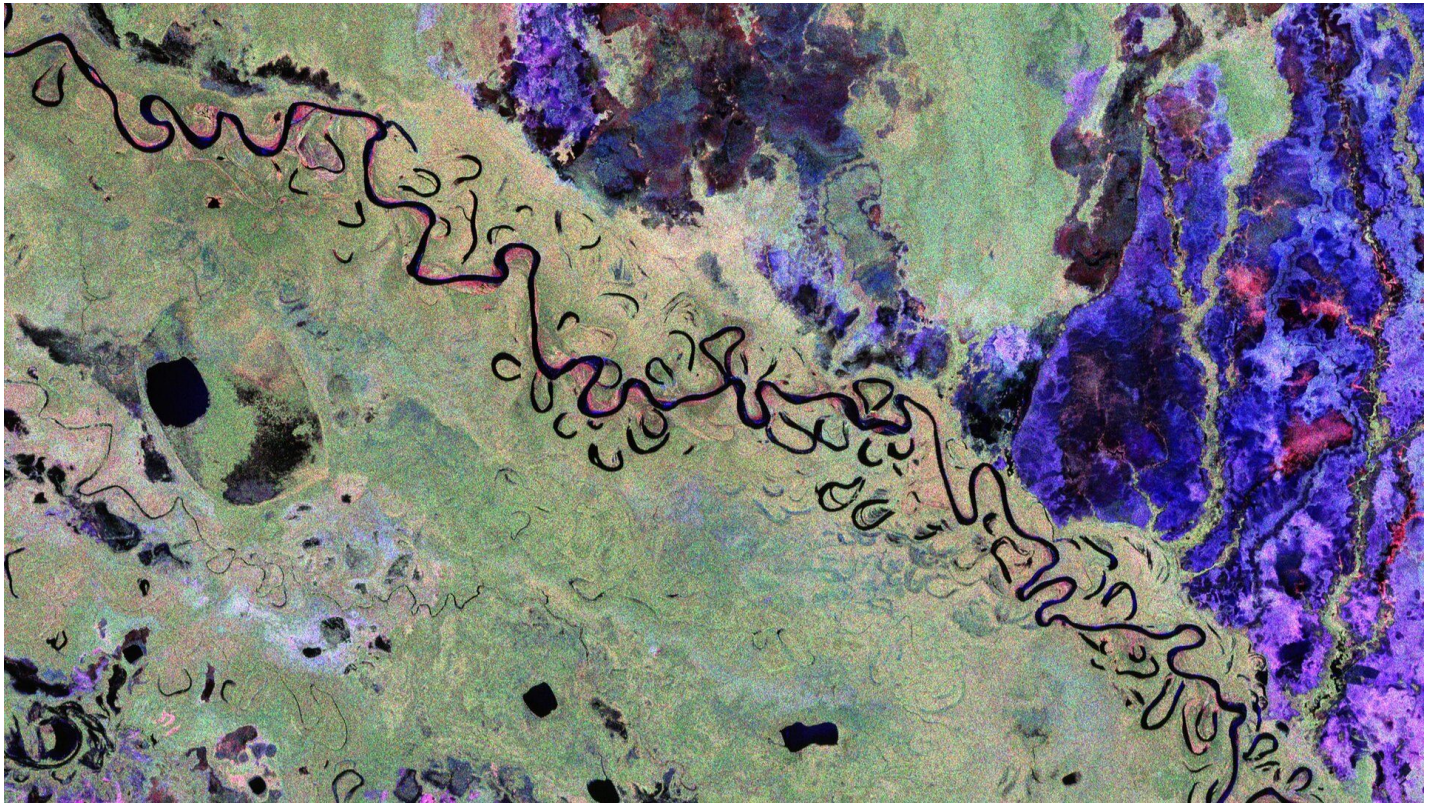




Bodesheim, P., Jung, M., Gans, F., Mahecha, M. D., & Reichstein, M. (2018). *Earth System Science Data*, *10*(3), 1327-1365.

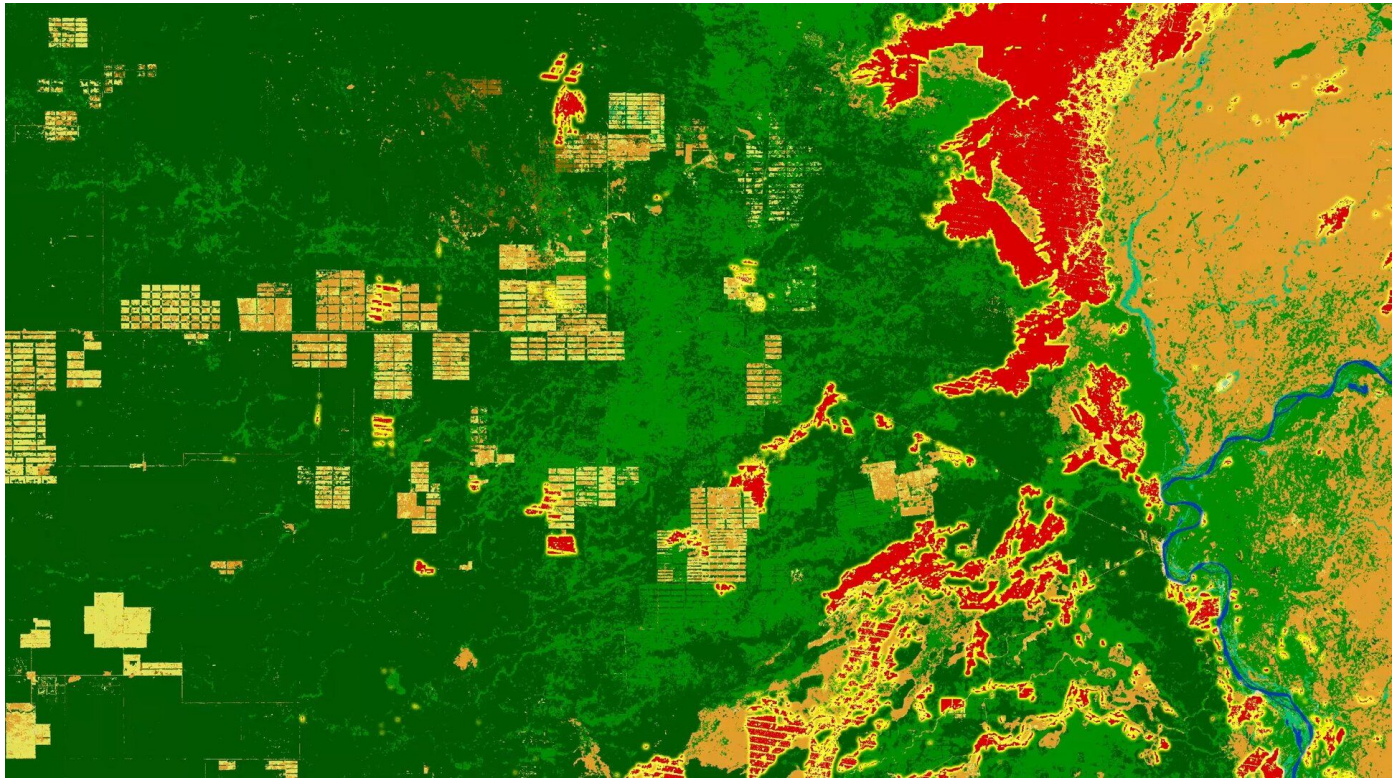
Unprecedented scales: Biomass

Biomass



Bolivian forest and landscape seen by ESA's Biomass satellite:
https://www.esa.int/Applications/Observing_the_Earth/FutureEO/Biomass/Biomass_satellite_returns_striking_first_images_of_forests_and_more

Fire-driven deforestation in the Gran Chaco



https://www.esa.int/Applications/Observing_the_Earth/Space_for_our_climate/Satellite_records_expose_fire_driving_Gran_Chaco_transformation

Other chances of a data-rich world



Winners of the Online Journalism Awards (OJAs) 2022: Ida Flik, André Pätzold, Benja Zehr
https://interaktiv.morgenpost.de/klimawandel-hitze-meeresspiegel-wassermangel-stuerme-unbewohnbar/

CONTENT & ORGANISATION

What you will be able to do after this module

- explain the key processes in the Earth system's components – atmosphere, hydrosphere, biosphere, geosphere and cryosphere – and how they interact
- quantify exchange fluxes of matter and energy between the spheres and close their budgets
- assess how global transformation processes, including human interventions, alter these processes
- judge which uncertainties remain in our process understanding and how they affect projections and models

Based on the module description 12-GEO-M-AG02 “Earth System Components”, M.Sc. Earth System Data Science and Remote Sensing (Leipzig University, as of 19 June 2025); learning outcomes translated from German.

Until December: atmosphere and water

Block	L	Date	Topic
Intro	**L01**	**13.10.2026**	**Course overview and organisation (today)**
I · Atmosphere	L02	20.10.2026	Atmosphere I: concepts and key variables
	L03	27.10.2026	Atmosphere II: radiation balance
	L04	03.11.2026	Atmosphere III: structure of the atmosphere
	L05	10.11.2026	Atmosphere IV: buoyancy forces
	L06	17.11.2026	Atmosphere V: circulation
II · Terrestrial hydrology	L07	24.11.2026	Terrestrial hydrology I: water-cycle concepts
	L08	01.12.2026	Terrestrial hydrology II
	L09	08.12.2026	Terrestrial hydrology II: measurements, Budyko framework

From mid-December: carbon, classification, a guest

Block	L	Date	Topic
III · Biogeochemistry	L10	15.12.2026	Biogeochemistry I: element cycles – the carbon cycle
	–	*22.12.–02.01.*	*Christmas break, no classes*
	L11	05.01.2027	Biogeochemistry II: terrestrial biogeochemistry
IV · Classification & special topics	L12	12.01.2027	Climate classification systems and their limitations
	L13	19.01.2027	Atmospheric rivers
	L14	26.01.2027	Forest damage and climate change – guest lecture: Prof. Henrik Hartmann
	L15	02.02.2027	Climate classifications and vegetation
	–	*Feb. 2027*	*Oral exams (dates to be confirmed)*

The formula sheet grows with the course: Parts I–IV

- One formula sheet for the whole module: constants, standard values and equations – always with units.
- It is released on Moodle part by part, matching the lecture blocks.
- Every equation is numbered (e.g. I-3) and tagged with the lecture in which it is used (e.g. L03).
- Consistent notation: one symbol = one quantity (e.g. H is only the sensible heat flux).
- The exercises use the same sheet – so does your exam preparation.

Part I
Atmosphere & radiation
from L02 · 20.10.2026

Part II
Terrestrial hydrology & Budyko
from L07 · 24.11.2026

Part III
Biogeochemistry
from L10 · 15.12.2026

Part IV
Climate classification
from L12 · 12.01.2027

How it fits together: lecture → exercise → assessment

Lecture	Self-study	Exercise	Assessment
Tuesdays · 15 sessions concepts & examples	slides, formula sheet, reading	weekly, starting today Python & Jupyter	portfolio + oral exam mid-February 2027

- Workload: 5 ECTS credits = 150 h in total = lecture 75 h (30 h in class + 45 h self-study) + exercise 75 h (15 h in class + 60 h self-study)
- That is about 10 h per week over 15 weeks – most of it outside the lecture hall
- After each lecture: go through the slides, look up the equations in the formula sheet, note your questions
- Your portfolio of class questions and exercises grows during the semester

Workload according to the module description 12-GEO-M-AG02 (Leipzig University, as of 19 June 2025).

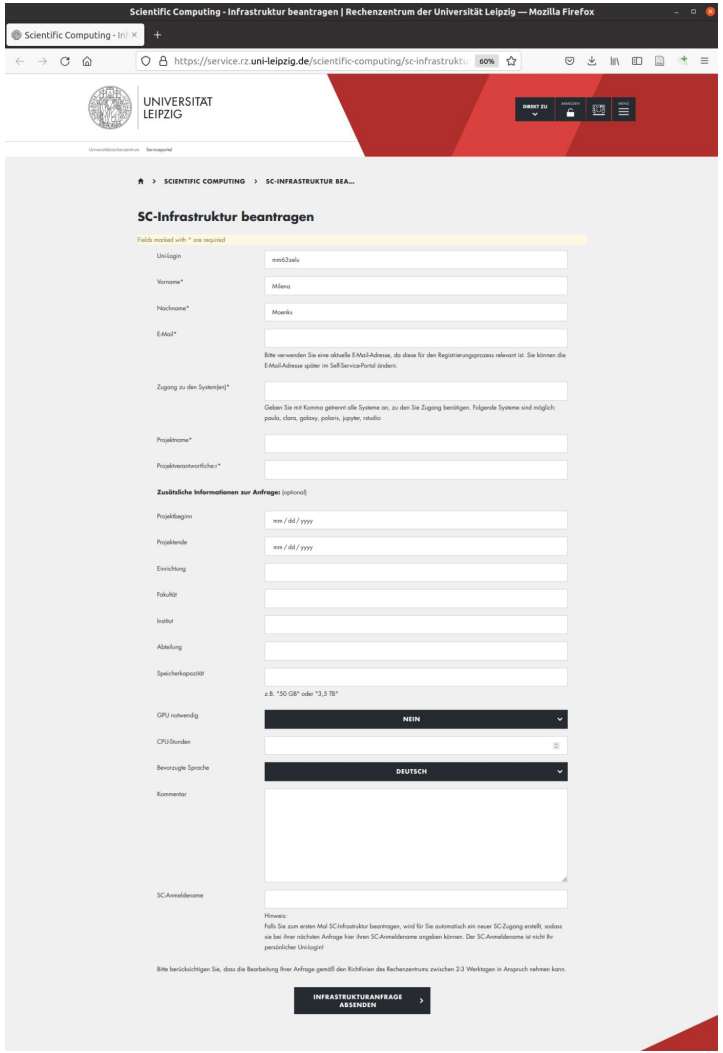
Exercises: apply today

Every week, starting today…

BUT: Please apply as soon as possible for access to the Jupyterlab from Scientific Computing (SC) because it can take a few days until your account is activated!

Follow the instructions on the official website:
https://www.sc.uni-leipzig.de/
> Create an account (if you are new)
> Manage account > Infrastruktur beantragen (if you already have an account)

You find more information on the SC resources in the Get Started document:
https://www.sc.uni-leipzig.de/user-doc/quickstart/



Your uni-login, first name, surname and corresponding email (e.g., uni mail address)

Zugang zu den System(en) ("access to systems"): Please enter *jupyter, paul, clara, paula*

Projektname: Master *Earth System Data Science*

Projektverantwortlicher: *Miguel Mahecha*

The other input fields are optional, please select your preferred language

When applying the first time, leave the field "SC-Anmeldename" blank

> After applying to the SC infrastructure you receive a mail, please confirm the request

Exercises: get familiar with Python and Jupyter

Every week, starting today with a short intro.

Until then, please get familiar with Python and Jupyter Notebook!
Super nerds → feel free to work in Julia

E.g. with these free YouTube tutorials:
https://www.youtube.com/watch?v=t8pPdKYpowI, or
https



Get an idea of what is a …

- Variable
- Strings
- Basic arithmetic and logic
- List
- Packages
- for loops, if statements
- How to plot a figure with matplotlib

Assessment: portfolio, then oral exam

Portfolio of class questions + exercises → needed for admission to the exam

Exam:

- Oral examination on the topics covered in the semester
- 15 min per person
- 1–2 days after the end of the lecture period, mid-February 2027 – dates to be confirmed

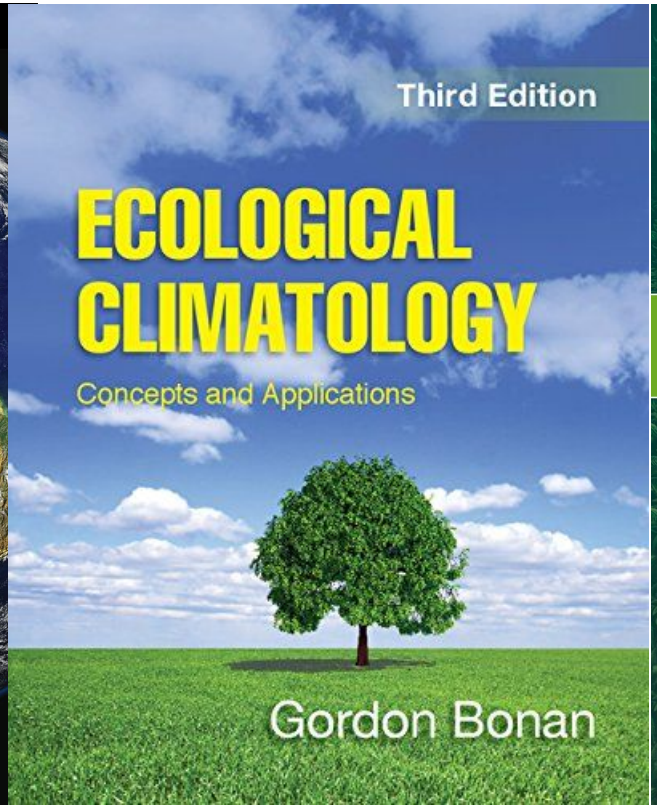
Communication: talk to us – email only in emergencies

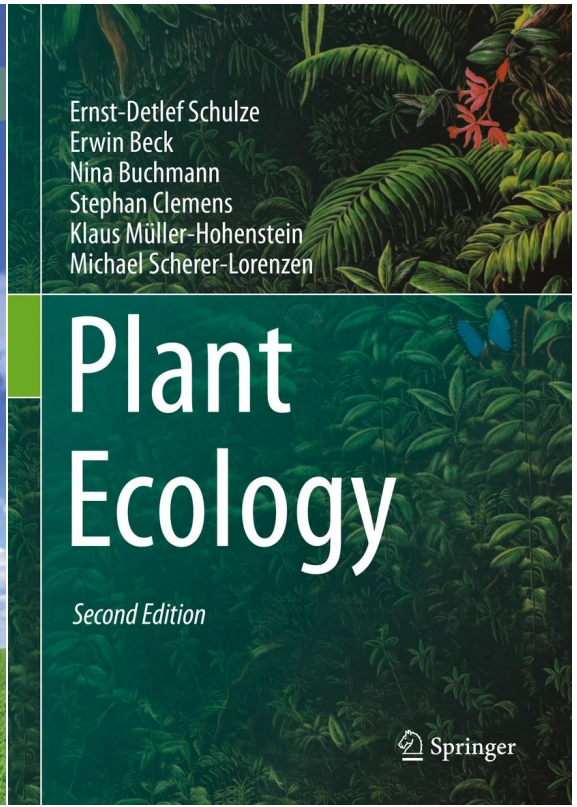
Emails only in an emergency:

- If so, with a clear subject line: "[M-AG02] Sick note exam".
- I receive messages via Moodle, but I cannot answer them.
- Sign emails with first and last name, add alias to name

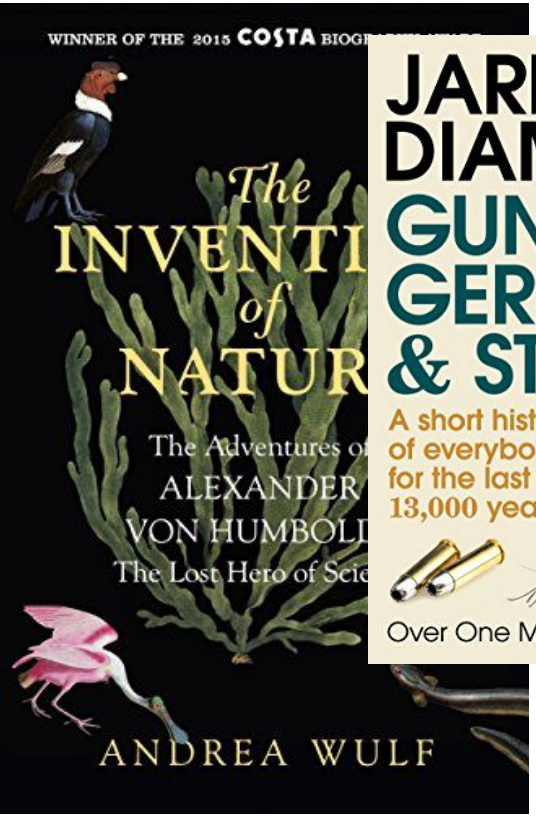
A few books that you could read

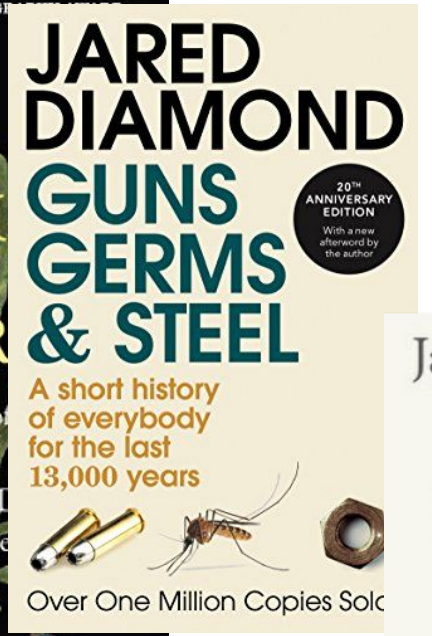


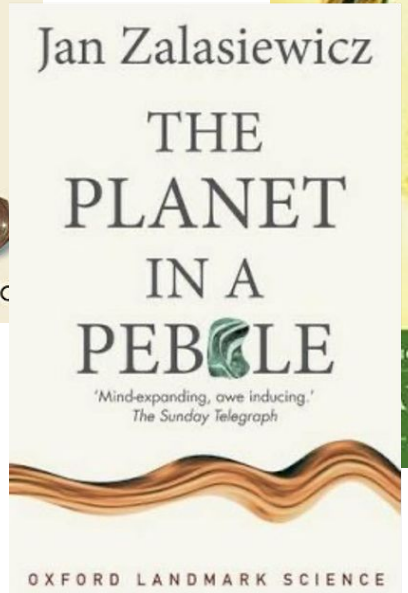


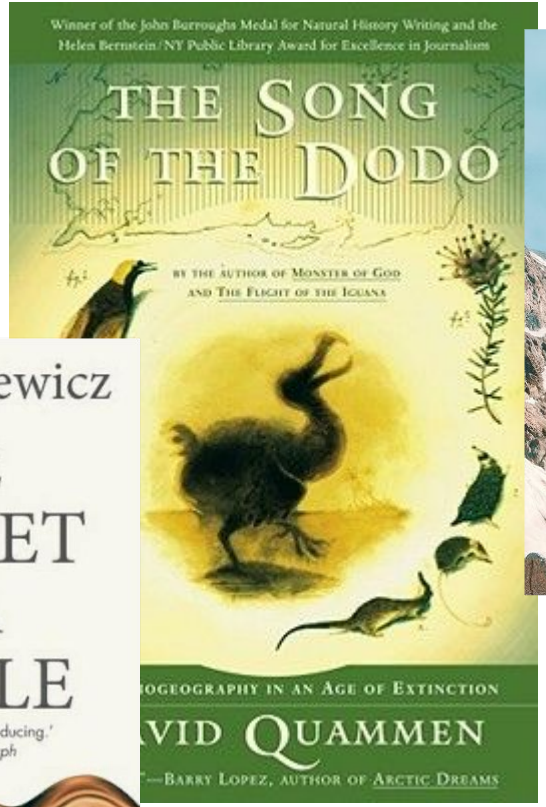


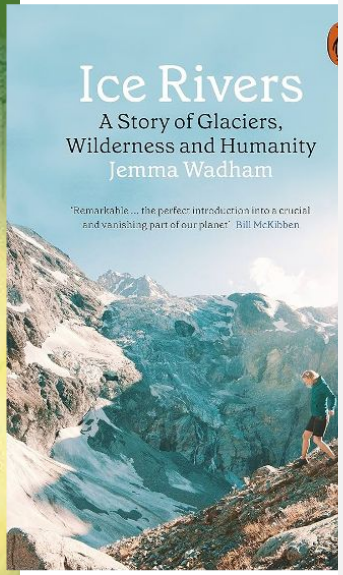
Read, read, read ...











… use the library



See you next week – today we continue with the exercise

- Today: first exercise – and apply for your JupyterLab account (Scientific Computing)
- Next week, L02 · 20.10.2026: Atmosphere I – concepts and key variables
- From L02 on Moodle: formula sheet Part I
- Questions? Ask right after the lecture or in the exercise